

RITVIJ NARAM

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MACHINE LEARNING SOFTWARE DEVELOPER | GENERATIVE AI

PROFESSIONAL SUMMARY

Information Systems graduate student and published researcher with 3+ years of experience in data analytics, SQL-based data engineering, ETL pipelines, and cloud platforms. Experienced in building data pipelines, dashboards, and AI-driven automation solutions, improving reporting efficiency by up to 97% and enabling data-driven decision-making. Hands-on ML/GenAI work includes TensorFlow/Keras, scikit-learn, LangChain/RAG, Gemini API, Docker, and Git/GitHub CI/CD.

EDUCATION

Master of Science, Information Systems | Northeastern University | Toronto, ON | December 2026

Bachelor of Engineering (BE), Computer Science | University of Pune | Pune, India | May 2021

CERTIFICATIONS

Certified Airtable Builder, Airtable, Canada | May 2025

CORE COMPETENCIES

Machine Learning & GenAI: Python | TensorFlow/Keras | scikit-learn | CNNs | LangChain | LLMs | RAG | Gemini API | Prompt Engineering | Embeddings | Model Evaluation

MLOps & Cloud Platforms: Docker | GitHub Actions CI/CD | Azure | AWS | GCP, Vertex AI, BigQuery & Kubernetes (current hands-on learning) | Model Monitoring

Data Analytics & Engineering: SQL | Python | Data Modeling | ETL Pipeline Development | Data Warehousing | Data Cleaning | Data Transformation | API Integration

Business Intelligence & Visualization: Tableau | Power BI | Dashboard Development | KPI Reporting | Data Storytelling | Business Performance Analytics

Cloud & Data Platforms: Microsoft Azure | Snowflake | Azure Functions | Cloud Data Pipelines | Serverless Architecture

AI & Automation: AI Chatbots | LangChain | LLM Integration | Process Automation | Workflow Optimization | NLP Applications

Database & Systems: Relational Databases | Database Design | Data Governance | Airtable | SharePoint | Enterprise Data Systems

Interpersonal: Cross-Functional Leadership | Problem Solving | Analytical Thinking | Requirements Gathering | Team Collaboration

Programming & Tools: Python | FastAPI | Git | REST APIs | Microsoft Excel | Advanced Excel Automation | Java | C++

PROFESSIONAL EXPERIENCE

Business Analytics & Systems Associate Co-op | Northeastern University | Toronto, ON | Jul 2025 - Dec 2025

● Led end-to-end campus analytics and systems initiatives, translating stakeholder requirements into production-ready solutions supporting thousands of users.

● Solely owned and delivered the Scorecard Data Project end to end, from stakeholder requirements and architecture through implementation, validation, documentation, deployment, and handoff; built scheduled Azure Functions ETL to pull Airtable/Asana API data, stage CSV/JSON checkpoints in Azure Blob Storage, and load curated Snowflake tables for Tableau reporting, improving data reliability by 50%.

● Secured the Scorecard pipeline with Azure Managed Identity, Key Vault, and environment variables; configured Snowflake warehouse/database/schema access and implemented incremental loads using monthly partitions and data hashes to prevent unnecessary reprocessing.

● Benchmarked pipeline performance and documented architecture, recovery procedures, and troubleshooting; standardized Git/GitHub workflows, GitHub Actions CI/CD, and Docker-based deployment practices for repeatable releases.

● Current technical upskilling: recreating ML deployment patterns on GCP with BigQuery, Vertex AI, Docker, and Kubernetes, including CI/CD, model monitoring, and container orchestration.

● Developed Python/Flask services and REST API integrations while maintaining mentorship-program databases and automation workflows supporting 300+ student and alumni records.

Software Developer | Mechanize Industries | Pune, IN | Sep 2022 - May 2024

● Built SQL-based ETL pipelines for weekly/monthly operational reporting, applying reusable validation and transformation logic to reduce manual processing and improve reporting consistency.

● Designed Power BI dashboards for 15+ KPIs and modeled OLTP operational data into analytics-ready structures, enabling 20+ stakeholders to monitor performance and investigate trends.

● Developed an AWS data-migration proof of concept using S3 for staged monthly data, RDS for transactional workloads, and Redshift for columnar analytics; applied partitioning and incremental-load checks to minimize redundant processing.

● Automated legacy Excel reporting with Python and SQL, reducing manual workload by ~30%; used Git-based code review and CI/CD practices to improve release consistency and maintainability.

● Partnered with business and technical stakeholders to validate datasets, benchmark pipeline performance, document architecture, and drive issues through production handoff.

Software Developer | Raja Industries | Pune, India | Aug 2021 - Aug 2022

● Developed an integrated reporting solution extending legacy operational systems with near-real-time analytics, covering requirements, database design, testing, deployment, and support.

● Built Python/SQL ingestion workflows and REST API integrations with incremental processing and data-quality checks to prepare operational data for downstream analytics.

● Prototyped AWS analytics architecture using S3 staging, RDS source data, and Redshift reporting models, separating transactional and analytical workloads for scalable querying.

● Containerized Python services with Docker and used Git/GitHub CI/CD practices for repeatable builds, testing, and deployments while optimizing complex SQL for reporting and audit requests.

● Produced technical documentation and user guides; collaborated with analysts and business users to troubleshoot issues, refine requirements, and improve system adoption.

RESEARCH PUBLICATIONS

Anomaly Detection | [Link](#) | July - 2021

Published in the International Research Journal of Modernisation in Engineering Technology and Science.

Artificial Intelligence System for Effective Detection of Animal Intervention in Croplands

| [Link](#) | Sept - 2021

Published in the International Journal for Research in Applied Science and Engineering Technology.

Wheeled Robot Development for Obstruction Avoidance Mechanism | [Link](#) | Sept - 2021

Published in the International Journal for Research in Applied Science and Engineering Technology.

PROJECTS

Anomaly Detection for Video Surveillance | Python, TensorFlow/Keras, OpenCV, CNN

● Built a CNN-based video anomaly-detection pipeline that preprocesses frames with OpenCV/NumPy, performs Keras/TensorFlow inference, and annotates abnormal events in video footage.

AI Chatbot for Student Support (Nora)

● Built LLM-based chatbot using LangChain and API integrations for automated student query handling and developed backend using Python and APIs.

● Improved response automation, reducing manual query handling and increasing response efficiency.

● Implemented retrieval-augmented generation (RAG) for context-aware responses.

Local Eco-Scanner | Kotlin, Jetpack Compose, CameraX, Gemini API | [GitHub](#)

● Developed a native Android application using Kotlin and Jetpack Compose, integrating CameraX and Gemini multimodal AI to identify household items and generate location-aware recycling guidance.

● Implemented camera lifecycle and permission handling, image rotation and downscaling, coroutine-based API requests, Canadian postal-code validation, error states, and private local-profile persistence.